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To the Editors,
Journal of Statistical Planning and Inference

Re: Ms. Ref. No. JSPI-D-26-00452 — “WAGER: Within-cell Antisymmetric Gain Evaluation of Resolution” — revised submission

Dear Editors,

Thank you for the opportunity to revise the manuscript above. This letter accompanies a substantially revised submission responding to the decision of major revision. Rather than a numbered point-by-point response, the sections below summarize the revision comprehensively; a table of the principal changes is given near the end of this letter.

The paper concerns the comparison of two probabilistic predictors when the labels are partly determined by a nuisance feature both models observe. In that situation the score difference confounds two quantities: a closer fit to the conditional label distribution of the nuisance feature, and genuinely better prediction of individual cases. Significance testing on the difference does not separate them, and the customary remedy — fitting a nuisance-only baseline and comparing against it — introduces estimation error, smoothing choices and model-family mismatch into what should be an exact statement about a fixed pair. Machine-learning benchmarks in computer vision supply the applications here, but the estimation problem is general.

WAGER answers that question directly, and the contribution is primarily methodological. Given two frozen probabilistic models, it forms their proper-score contrast, matches test examples sharing a declared prior feature, and crosses their labels. This yields an exact finite-sample identity — the observed gain equals a prior-transported gain plus an antisymmetric instance-alignment gain — whose residual is an order-two U-statistic that, under the quadratic score, equals the within-cell prediction–label covariance the new model has gained, linking it to the classical resolution term of proper-score decomposition. The estimator runs in $O(NK)$ time on cached predictions, fits no baseline model, has no tunable hyperparameter, and comes with cluster-robust intervals and a cell-stratified randomization test. This revision adds four further results aimed at this journal’s readership specifically: the covariance identity is shown to hold for the entire Bregman-score family, not the quadratic score alone; an asymptotic normality theorem justifies the reported intervals in the large-sample limit; a Cauchy–Schwarz sensitivity bound quantifies how far an unrecorded confounder could move the residual from what a fully adjusted audit would find; and, at a trivial audit feature, the statistic and its interval reduce exactly to a clustered Diebold–Mariano/Giacomini–White test, situating the estimator precisely relative to comparative forecast evaluation. Two further exact results were already established: leave-one-out transport removes an attenuation bias of $(n_c - 1)/n_c$ that a naive in-sample plug-in incurs, and coarsening the audit cell shifts the estimate by an explicit between-cell covariance term.

Because the method is a measuring instrument, the paper takes its validation seriously. Beyond recovery and null designs, we verify the coverage of the primary interval in a regime that stresses it, and we report discriminant-validity experiments establishing what the two channels do and do not separate — including an honest limitation, that a monotone recalibration moves score mass between the channels, together with the calibration-matched protocol that removes it and

a demonstration that the protocol works.

The empirical results show the instrument earning its keep, and the clearest demonstration concerns a method the field already compares rather than any predictor of ours. We evaluate the released causal MOTIFS checkpoint of Tang et al. (CVPR 2020) in both of its modes with the authors’ own code on the standard Visual Genome PredCls split, reproducing their published behaviour, and decompose the debiased model against the biased baseline. Read naively, the Total Direct Effect variant appears to lose a little instance alignment. That reading does not survive the control this paper argues for: the two models differ sharply in confidence, and once temperatures are matched on held-out images the proper-score difference is overwhelmingly prior-transported under both the quadratic and the log score, though the two scores disagree on whether any instance-alignment remainder survives (-0.00006 , 95% CI $[-0.00120, +0.00109]$ under the quadratic score; $+0.04396$, CI $[+0.04079, +0.04714]$ under the log score) — in either reading at least an order of magnitude smaller than the prior-channel shift. A widely used debiasing method therefore buys ten points of mean recall while changing within-cell instance alignment by at most a small fraction of that amount, possibly none — which is what its construction subtracts, so the finding characterises the method rather than indicting it, while showing that a metric read as evidence about rare predicates can move sharply on evidence that is mostly, and perhaps entirely, of a different kind.

Two further results point in complementary directions. A model reading real pixels through a frozen CLIP encoder scores 0.04655 *lower* overall — an evaluator reading that number would discard it — yet aligns with individual instances significantly *better* than the model that beats it, its entire deficit lying in the annotation prior and its advantage surviving both post-hoc prior correction and calibration matching. Applied unchanged to long-tailed image classification across seeds and imbalance ratios, a calibration-matched reanalysis overturns what the raw decomposition appears to say about two re-weighting schedules. A third domain with neither a spatial nor a class-frequency prior — long-tailed text classification — shows the same estimator, unmodified, produces a different and equally interpretable composition: the schedule that buys real instance alignment on one domain buys almost none on the other, which is itself evidence that the decomposition tracks a genuine property of the training-data pair rather than a fixed property of a method’s name.

I believe the work fits this journal’s remit in the theory and practice of statistical inference and model comparison: the central contribution is an estimand and its exact finite-sample and asymptotic inference theory — U-statistic identities, a Hájek influence function, an asymptotic normality result, and a sensitivity analysis for an unmeasured confounder in the tradition of Rosenbaum and Cinelli & Hazlett — together with an exact relationship to the classical Diebold–Mariano/Giacomini–White tests for comparative forecast evaluation. The computer-vision and text-classification benchmarks are the applications on which the instrument is exercised and validated, not the claim itself; no state-of-the-art result is asserted, and none is needed for the argument.

Summary of principal changes in this revision.

- Four new theoretical results aimed at this journal’s readership: the Bregman-score generalization of the covariance identity (Theorem 2), the asymptotic normality theorem for the dataset-level estimator (Theorem 3), the sensitivity bound for an unrecorded confounder and its data-free worst-case corollary (Proposition 3 and Corollary), and the exact reduction to a clustered Diebold–Mariano/Giacomini–White test at a trivial audit feature (Corollary).
- A third empirical domain, long-tailed text classification on 20 Newsgroups, with neither a spatial nor a class-frequency prior, added to the existing scene-graph and long-tailed

image classification studies.

- Two results corrected after further internal review, both verified by hand before editing: a related-work argument claiming a self-prediction bias "does not cancel under subtraction" was found to be false when both compared models share the same audit partition, and has been replaced with the paper's actually-valid distinguishing point (three newly verified citations added); and the new asymptotic normality theorem's proof required an explicit condition, now stated, that the paper's own heavy-tailed real-data application does not attain in-sample, with the reported coverage simulation supplying the finite-sample evidence in its place.
- The headline empirical finding (Section 4.6) has been restated to reflect a robustness check the manuscript itself reports: the quadratic and log scores agree that the audited debiasing method's improvement is overwhelmingly prior-transported, but disagree on whether any instance-alignment remainder is exactly zero, and the revision now states this precisely rather than reporting only the score that read as null.

All code, cached model outputs and scripts needed to reproduce every number, table and figure are publicly available at <https://github.com/vinhqdang/WAGER>, including a verification script that checks each reported value against the committed results files.

The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. I am the sole author and approve its submission. I declare no competing financial or personal interests, and the work received no specific grant from any funding agency. In accordance with the journal's policy, the manuscript includes a declaration of generative AI use in its preparation.

Thank you for your consideration.

Yours sincerely,

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